

Vinh Ly

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Summary

Software Engineer with **18 years** of end-to-end ownership of global video streaming, real-time auction, and large-scale ad-serving platforms. Designed and operated services on **AWS**, **Kubernetes**, **Terraform**, **Go**, and **C++**. Delivered a 27% latency reduction while handling 1.2 billion daily requests, collaborating with product, security, and data-science partners. Current focus is building hardened edge compute for air-gapped and isolated network environments. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Cloud & Infrastructure: AWS, Terraform, Kubernetes, Linux, GCP

Programming Languages: Go, C++, Python, Java, Bash, TypeScript, JavaScript, HTML, CSS, SQL

Container & Orchestration: Docker, Kubernetes, Helm, Istio, Podman

Automation & CI/CD: Git, Jenkins, Spinnaker, Argo CD, Make, Prometheus, Grafana, Ceph

Work Experience

Staff Software Engineer, YouTube (02/2025 - Present), Remote

- Re-architected the live-stream ingest pipeline on **AWS** and **Kubernetes**, cutting processing latency by 27% and supporting 1.2 billion daily requests.
- Partnered with security and data-science teams to embed real-time anomaly detection, preventing 12 million potential abuse events.
- Led on-call incident response for the video delivery stack, decreasing mean-time-to-recovery by 35% through automated diagnostics.
- Integrated **Go** microservices with existing C++ transcoding back-end, improving throughput by 18% while maintaining codec compatibility.
- Implemented Terraform modules for multi-region deployment, reducing provisioning time from days to under two hours.

Staff Software Engineer, Whatnot (05/2024 - 02/2025), Remote

- Built a Kubernetes-based auction engine on **AWS**, scaling to 250 k concurrent users with sub-second bid latency.
- Collaborated with product managers to launch a new “watch-now” feature, driving a 15% increase in user engagement.
- Created Terraform-driven infrastructure for isolated test environments, cutting environment spin-up cost by 40%.
- Authored Go services for real-time inventory sync, increasing item availability accuracy by 22%.
- Managed on-call rotation for the marketplace platform, achieving a 99.98% uptime SLA.

Staff Software Engineer/Manager, Google (03/2015 - 04/2024), Remote

- Directed a team of 12 engineers delivering a global ad-serving platform on **GCP** and **Kubernetes**, handling 3 billion daily impressions.

- Introduced Go-based latency monitoring agents, uncovering bottlenecks that shaved 12 ms off page load times.
- Implemented automated canary deployments with Spinnaker, decreasing release failures from 4% to 0.5%.
- Coordinated cross-functional on-call duties, achieving a 30% reduction in critical incident escalation time.
- Standardized Terraform infrastructure across data centers, reducing configuration drift by 90%.

Software Engineer, Google (02/2011 - 02/2015), Remote

- Automated build pipelines with Jenkins, cutting nightly build duration from 90 minutes to 45 minutes.
- Ported key components to **Go**, reducing codebase size by 15% and simplifying maintenance.
- Developed C++ core services for search indexing, improving index refresh rate by 20%.
- Supported on-call rotation for the search backend, maintaining 99.9% availability.

Software Design Engineer II, Microsoft (10/2007 - 12/2011), Remote

- Engineered **Python** automation scripts for Azure resource provisioning, accelerating test environment setup by 50%.
- Contributed to the Windows kernel networking stack, enhancing packet processing throughput by 10%.
- Built CI pipelines with Team Foundation Server, reducing integration time for feature branches.
- Participated in on-call support for enterprise cloud services, achieving a 98.7% SLA compliance.

Software Design Engineer Intern, Microsoft (06/2006 - 08/2006), Remote

- Implemented a **Python** prototype for automated log analysis, decreasing manual review effort by 70%.
- Collaborated with senior engineers to refactor legacy C++ modules, improving code readability.

Education

B.S. Honors Computer Science

Virginia Tech | 2004 - 2007